# MARKING SCHEME

## T.I.SCHOOL
## Ambattur, Chennai-53

### Grade X – Artificial Intelligence – Periodic Test 2
### PT2 X Sep 2627 Scoring Key

Time: 2 Hours    |    Maximum Marks: 50
For Academic Evaluation & Student Reference

| Component | Question Pattern | Marks |
|---|---|---|
| Section A | Objective – Q1 to Q5 | 24 |
| Section B-I | Employability Skills – answer 3 of 5 | 6 |
| Section B-II | AI Evaluation & Data – answer 4 of 6 | 8 |
| Section B-III | Applied / Case-Based – answer 3 of 5 | 12 |
| **Total** | **50 marks** | 50 |

## GENERAL EVALUATION GUIDELINES

1. Evaluate answers strictly with reference to the value points and marking guidance provided in this scheme.

2. The suggested answers are value points, not the only acceptable wording. Award marks for any correct, relevant and conceptually equivalent response.

3. For objective questions, award 1 mark only for the correct option. No negative marking is applicable.

4. For subjective questions, award marks independently for each relevant value point. Do not penalise a student merely for using different wording.

5. Do not award more than the maximum marks prescribed for a question.

6. Where a question asks for two points, examples or practices, each distinct correct/relevant point normally carries 1 mark unless a specific breakup is provided.

7. For numerical questions, award method/working marks where the prescribed breakup is satisfied, even if a minor arithmetic error occurs after a correct method.

8. If an answer contains an extra point that is incorrect but the required correct points are clearly present, assess the required correct points without exceeding the maximum marks.

9. Check that all attempted questions are assessed and that question-wise marks are correctly totalled and transferred.

# SECTION A – OBJECTIVE TYPE QUESTIONS

Maximum Marks: 24 | Each correct response carries 1 mark.

## Question 1 | Answer any 4 out of the given 6 questions

| | | |
|---|---|---|
| (i) | A student listens carefully to a teacher, asks relevant questions, and repeats the main points to confirm understanding. Which communication skill is being demonstrated?<br><br>A. Passive listening<br>B. Active listening<br>C. Written communication<br>D. Visual communication | Answer:<br>B. Active listening<br><br>1 mark |
| (ii) | Ravi has several assignments due next week. He prepares a timetable and completes the most urgent tasks first. Which self-management skill is Ravi using?<br><br>A. Stress management<br>B. Goal setting<br>C. Time management<br>D. Self-motivation | Answer:<br>C. Time management<br><br>1 mark |
| (iii) | Which practice is the most appropriate when creating a password for an online account?<br><br>A. Using your date of birth<br>B. Using your name and phone number<br>C. Using a strong combination of letters, numbers, and symbols<br>D. Using the same password for every account | Answer:<br>C. Using a strong combination of letters, numbers, and symbols<br><br>1 mark |
| (iv) | Meena notices that students in her school struggle to find affordable stationery. She develops a low-cost stationery delivery service to solve this problem. Which entrepreneurial quality is mainly demonstrated?<br><br>A. Identifying an opportunity<br>B. Avoiding risk<br>C. Following instructions<br>D. Managing stress | Answer:<br>A. Identifying an opportunity<br><br>1 mark |
| (v) | A school replaces disposable plastic bottles with reusable water bottles and installs systems to reduce water wastage. What is the main objective of these practices?<br><br>A. Increasing consumption<br>B. Promoting sustainable development<br>C. Increasing electricity usage<br>D. Reducing student participation | Answer:<br>B. Promoting sustainable development<br><br>1 mark |
| (vi) | A student wants to start an eco-friendly product business. She researches customer needs online, plans her time, communicates with customers effectively, and uses recycled materials. Which statement best describes this approach?<br><br>A. She is using only ICT skills.<br>B. She is using only entrepreneurial skills.<br>C. She is combining multiple employability skills to achieve her goal.<br>D. She is focusing only on green skills. | Answer:<br>C. She is combining multiple employability skills to achieve her goal.<br><br>1 mark |

## Question 2 | Answer any 5 out of the given 6 questions

| | | |
|---|---|---|
| (i) | During a group discussion, a student listens carefully to others, waits for their turn, and responds appropriately. Which skill is being demonstrated?<br><br>A. Aggressive communication<br>B. Active listening<br>C. Time management<br>D. Self-motivation | Answer:<br>B. Active listening<br><br>1 mark |
| (ii) | A student has an examination, project submission, and presentation scheduled in the same week. She prepares a schedule and prioritises tasks according to their deadlines. Which skill is she demonstrating?<br><br>A. Stress management<br>B. Creative thinking<br>C. Time management<br>D. Leadership | Answer:<br>C. Time management<br><br>1 mark |
| (iii) | Which of the following is the best practice for protecting personal information online?<br><br>A. Sharing passwords with close friends<br>B. Using the same password for every account<br>C. Using strong passwords and enabling security features<br>D. Clicking links received from unknown sources | Answer:<br>C. Using strong passwords and enabling security features<br><br>1 mark |
| (iv) | A student starts a business selling products made from recycled materials. Which combination of skills is most clearly demonstrated?<br><br>A. Entrepreneurial and Green Skills<br>B. Communication and ICT Skills only<br>C. Self-management and communication only<br>D. Mathematical and programming skills | Answer:<br>A. Entrepreneurial and Green Skills<br><br>1 mark |
| (v) | An AI model correctly predicts whether an email is spam or not spam for 90 out of 100 emails. What is the accuracy of the model?<br><br>A. 9%<br>B. 10%<br>C. 90%<br>D. 100% | Answer:<br>C. 90%<br><br>1 mark |
| (vi) | A dataset contains the following values: 10, 20, 20, 30, 40. What is the mean of the dataset?<br><br>A. 20<br>B. 24<br>C. 25<br>D. 30 | Answer:<br>B. 24<br><br>1 mark |

## Question 3 | Answer any 5 out of the given 6 questions

| | | |
|---|---|---|
| (i) | Which metric measures the proportion of correct predictions out of the total number of predictions made by an AI model?<br><br>A. Precision<br>B. Recall<br>C. Accuracy<br>D. F1 Score | Answer:<br>C. Accuracy<br><br>1 mark |
| (ii) | An AI model identifies 80 positive cases correctly out of 100 actual positive cases. What is the recall of the model?<br><br>A. 20%<br>B. 40%<br>C. 80%<br>D. 100% | Answer:<br>C. 80%<br><br>1 mark |
| (iii) | In a classification problem, the model predicts that a student is “At Risk”, and the student is actually “At Risk.” This is an example of:<br><br>A. True Positive<br>B. False Positive<br>C. True Negative<br>D. False Negative | Answer:<br>A. True Positive<br><br>1 mark |
| (iv) | Consider the dataset: 5, 10, 15, 20, 25. What is the median of the dataset?<br><br>A. 10<br>B. 15<br>C. 18<br>D. 20 | Answer:<br>B. 15<br><br>1 mark |
| (v) | Which statistical measure represents the value that occurs most frequently in a dataset?<br><br>A. Mean<br>B. Median<br>C. Mode<br>D. Range | Answer:<br>C. Mode<br><br>1 mark |
| (vi) | A dataset contains the values: 12, 15, 18, 20, 25. What is the range of the dataset?<br><br>A. 10<br>B. 12<br>C. 13<br>D. 15 | Answer:<br>C. 13<br><br>1 mark |

## Question 4 | Answer any 5 out of the given 6 questions

| | | |
|---|---|---|
| (i) | A school develops an AI model to identify students who may need additional academic support. The model identifies 50 students as “at risk,” but only 40 of them actually need support. What does this situation mainly help the school evaluate?<br><br>A. How many actual at-risk students were missed<br>B. How accurately the model's positive predictions identify at-risk students<br>C. How many students are present in the entire school<br>D. How quickly the model processes the data | Answer:<br>B. How accurately the model's positive predictions identify at-risk students<br><br>1 mark |
| (ii) | An AI model is tested on 1,000 images. It achieves 95% accuracy, but the dataset contains 950 normal images and only 50 defective images. Why might accuracy alone be misleading in this situation?<br><br>A. Accuracy cannot be calculated for image data.<br>B. The model may perform poorly on the minority class despite having high overall accuracy.<br>C. Accuracy is always lower when datasets are large.<br>D. Accuracy can only be used for numerical data. | Answer:<br>B. The model may perform poorly on the minority class despite having high overall accuracy.<br><br>1 mark |
| (iii) | A medical AI system is designed to identify patients who may have a serious disease. The developers want to minimise the number of actual patients with the disease who are incorrectly classified as healthy. Which evaluation measure should they focus on?<br><br>A. Recall<br>B. Accuracy<br>C. Mean<br>D. Range | Answer:<br>A. Recall<br><br>1 mark |
| (iv) | Two AI training datasets have the following average temperatures: Dataset A: Mean = 25°C; Dataset B: Mean = 25°C. However, Dataset A has values clustered closely around 25°C, while Dataset B contains values ranging from 10°C to 40°C. What can you conclude?<br><br>A. Both datasets have exactly the same distribution.<br>B. Dataset B has greater variability than Dataset A.<br>C. Dataset A has a higher mean.<br>D. Dataset B has no variation. | Answer:<br>B. Dataset B has greater variability than Dataset A.<br><br>1 mark |
| (v) | An AI system uses the ages of students in a dataset. Most students are between 14 and 16 years old, but one entry is recorded as 95 years old. Which statistical measure would be most affected by this unusual value?<br><br>A. Median<br>B. Mean<br>C. Mode<br>D. Frequency | Answer:<br>B. Mean<br><br>1 mark |
| (vi) | A student analyses the following test scores: 40, 42, 43, 44, 91. The student says, “The typical score is around 52.” Which concept should make the student question this conclusion?<br><br>A. The unusually high value affects the mean.<br>B. The mode is always greater than the mean.<br>C. The range does not depend on extreme values.<br>D. The median is always equal to the mean. | Answer:<br>A. The unusually high value affects the mean.<br><br>1 mark |

## Question 5 | Answer any 5 out of the given 6 questions

| | | |
|---|---|---|
| (i) | A school develops an AI model to detect students who are likely to drop out. During testing, the model correctly identifies 90 students who are actually at risk, but fails to identify 10 students who are also actually at risk. What should the school focus on if its priority is to identify as many at-risk students as possible?<br><br>A. Precision<br>B. Recall<br>C. Mean<br>D. Range | Answer:<br>B. Recall<br><br>1 mark |
| (ii) | An AI-based security system identifies 100 people as suspicious. After checking, only 70 of them are actually suspicious. The security team wants to know how reliable the system's positive predictions are. Which metric should they examine?<br><br>A. Recall<br>B. Precision<br>C. Accuracy<br>D. Median | Answer:<br>B. Precision<br><br>1 mark |
| (iii) | A hospital tests an AI model that predicts whether a patient has a particular disease. Out of 200 patients, the model correctly classifies 180 patients. However, doctors discover that the model is particularly poor at identifying patients who actually have the disease. What is the most appropriate next step?<br><br>A. Rely only on the 90% accuracy score.<br>B. Evaluate recall to determine how many actual positive cases were detected.<br>C. Remove all positive cases from the dataset.<br>D. Increase the number of decimal places in the accuracy value. | Answer:<br>B. Evaluate recall to determine how many actual positive cases were detected.<br><br>1 mark |
| (iv) | A fitness application collects the daily step counts of five students: 4,000, 5,000, 5,500, 6,000, 25,000. The developer wants to represent the typical daily activity level without allowing the unusually high value to distort the result too much. Which measure would be most appropriate?<br><br>A. Mean<br>B. Median<br>C. Range<br>D. Maximum | Answer:<br>B. Median<br><br>1 mark |
| (v) | An AI system is trained using the following monthly electricity consumption data from a classroom: 100, 105, 110, 108, 107, 500. The developer notices that 500 units is much higher than the other observations. What should the developer do before using this data for analysis?<br><br>A. Automatically delete the value.<br>B. Ignore the value because it is inconvenient.<br>C. Investigate whether the value is a genuine observation or an error.<br>D. Replace the value with zero. | Answer:<br>C. Investigate whether the value is a genuine observation or an error.<br><br>1 mark |
| (vi) | A school wants to build an AI model to predict student performance. The dataset contains marks, attendance, study hours, and student names. The data scientist decides not to use student names as a feature for prediction. Why is this decision appropriate?<br><br>A. Names are always numerical values.<br>B. Names are unlikely to provide meaningful predictive information and may introduce irrelevant information.<br>C. AI models cannot process more than three features.<br>D. Student names determine the mean of the dataset. | Answer:<br>B. Names are unlikely to provide meaningful predictive information and may introduce irrelevant information.<br><br>1 mark |

# SECTION B-I – Employability Skills

Answer any 3 out of the given 5 questions. Answer each question in 20–30 words.

| Q. | QUESTION / SUGGESTED VALUE POINTS | MARKING GUIDANCE |
|---|---|---|
| 1 | Why is active listening important for effective communication? Give one example of how a student can demonstrate active listening.<br><br>Suggested value points:<br>• Active listening helps understand the speaker correctly.<br>• It reduces misunderstandings and builds trust.<br>• Example: listening attentively and asking relevant questions. | 1 mark for the im portance/benefit; 1 mark for a relevant example of active listening.<br><br>Maximum: 2 marks |

| Q. | QUESTION / SUGGESTED VALUE POINTS | MARKING GUIDANCE |
|---|---|---|
| 2 | How can setting clear goals and managing time effectively help a student complete academic tasks successfully?<br><br>Suggested value points:<br>• Clear goals provide direction.<br>• Effective time management helps prioritise tasks and meet deadlines.<br>• It can reduce stress and support systematic completion of academic work. | Award 1 mark for clear goals/direction; 1 mark for time-ma nagement benefits such as prioritisation, deadlines or reduced stress.<br><br>Maximum: 2 marks |

| Q. | QUESTION / SUGGESTED VALUE POINTS | MARKING GUIDANCE |
|---|---|---|
| 3 | Mention two safe practices students should follow while using the internet to protect their personal information and online accounts.<br><br>Suggested value points:<br>• Use strong, unique passwords.<br>• Avoid sharing personal information online.<br>• Enable two-factor authentication.<br>• Avoid suspicious links. | 1 mark each for any two correct and relevant safe practices.<br><br>Maximum: 2 marks |

| Q. | QUESTION / SUGGESTED VALUE POINTS | MARKING GUIDANCE |
|---|---|---|
| 4 | How can identifying a problem in the community help a person develop a successful business idea?<br><br>Suggested value points:<br>• A community problem reveals people's needs.<br>• A useful product/service can be developed to solve the problem.<br>• Solving the problem can create business value. | 1 mark for identifying a need/problem as an opportunity; 1 mark for linking it to a useful product/service and business value.<br><br>Maximum: 2 marks |

| Q. | QUESTION / SUGGESTED VALUE POINTS | MARKING GUIDANCE |
|---|---|---|
| 5 | Suggest two practical actions students can take in school to conserve resources and promote environmental sustainability.<br><br>Suggested value points:<br>• Reduce paper usage by using digital resources.<br>• Conserve electricity by switching off lights/devices when not required. | 1 mark each for any two practical, relevant sustainability actions.<br><br>Maximum: 2 marks |

# SECTION B-II – AI Evaluation & Data

Answer any 4 out of the given 6 questions in 20–30 words each.

| Q. | QUESTION / SUGGESTED VALUE POINTS | MARKING GUIDANCE |
|---|---|---|
| 1 | Why is it important to evaluate an AI model before using it in a real-world situation?<br><br>Suggested value points:<br>• Determine accuracy and reliability.<br>• Identify errors and weaknesses.<br>• Check whether predictions are suitable for real-world use. | Award 1 mark for accuracy/reliability and 1 mark for identifying weaknesses/suitability for real-world use.<br><br>Maximum: 2 marks |

| Q. | QUESTION / SUGGESTED VALUE POINTS | MARKING GUIDANCE |
|---|---|---|
| 2 | A model has high accuracy but performs poorly when identifying a particular class. Why can accuracy alone be misleading?<br><br>Suggested value points:<br>• Accuracy measures overall correct predictions.<br>• If one class dominates the dataset, high accuracy may hide poor performance on the less-represented class. | 1 mark for explaining overall accuracy; 1 mark for explaining class imbalance/minority-class weakness.<br><br>Maximum: 2 marks |

| Q. | QUESTION / SUGGESTED VALUE POINTS | MARKING GUIDANCE |
|---|---|---|
| 3 | What is the difference between a True Positive and a False Positive? Give a simple example.<br><br>Suggested value points:<br>• True Positive: the model correctly predicts a positive case.<br>• False Positive: the model predicts positive when the actual case is negative.<br>• Example: wrongly identifying a healthy person as sick is a false positive. | 1 mark for correct TP definition; 1 mark for correct FP definition with a relevant example.<br><br>Maximum: 2 marks |

| Q. | QUESTION / SUGGESTED VALUE POINTS | MARKING GUIDANCE |
|---|---|---|
| 4 | A dataset contains extreme values. How can such values affect the mean, and why might the median be preferred?<br><br>Suggested value points:<br>• Extreme values can pull the mean significantly upward or downward.<br>• The median is less affected by outliers and may better represent the typical value. | 1 mark for effect on mean; 1 mark for why median may be preferred.<br><br>Maximum: 2 marks |

| Q. | QUESTION / SUGGESTED VALUE POINTS | MARKING GUIDANCE |
|---|---|---|
| 5 | A student records the marks 20, 25, 25, 30, and 50. What is the mode, and what does it tell us?<br><br>Suggested value points:<br>• Mode = 25.<br>• 25 occurs most frequently.<br>• It represents the most commonly occurring mark. | 1 mark for correct mode; 1 mark for correct interpretation.<br><br>Maximum: 2 marks |

| Q. | QUESTION / SUGGESTED VALUE POINTS | MARKING GUIDANCE |
|---|---|---|
| 6 | Why is understanding the distribution and variability of data important when preparing data for an AI model?<br><br>Suggested value points:<br>• It helps identify patterns and unusual values.<br>• It reveals differences/variability in data.<br>• This supports better preprocessing and more reliable predictions. | 1 mark for identifying patterns/unusual values; 1 mark for linking distribution/variability to preprocessing or reliable predictions.<br><br>Maximum: 2 marks |

## SECTION B-III – Applied / Case-Based Questions

Answer any 3 out of the given 5 questions in 50–80 words each.

| Q. | QUESTION / SUGGESTED VALUE POINTS | MARKING GUIDANCE |
|---|---|---|
| 1 | During a group project, Riya frequently interrupts her teammates and dismisses their suggestions without listening. The team is unable to reach an agreement. What communication problems are present, and what should Riya do to improve communication within the team?<br><br>Suggested value points:<br>• Identify poor listening/interruption.<br>• Recognise lack of respect for different opinions and resulting misunderstandings/teamwork problems.<br>• Practise active listening: allow others to finish, ask clarification questions and acknowledge ideas.<br>• Communicate respectfully, consider viewpoints and contribute without dismissing others. | 1 mark: identifies communication problems; 1 mark: explains impact; 1 mark: suggests active-listening actions; 1 mark: suggests respectful, collaborative communication.<br><br>Maximum: 4 marks |

| Q. | QUESTION / SUGGESTED VALUE POINTS | MARKING GUIDANCE |
|---|---|---|
| 2 | Arun has three assignments, a practical examination, and a project submission scheduled in the same week. He feels stressed and keeps postponing his work. Suggest a practical strategy using goal setting, time management, and stress management to help Arun complete his responsibilities effectively.<br><br>Suggested value points:<br>• List tasks and set clear, achievable daily goals.<br>• Prioritise by deadlines and divide large tasks into smaller activities.<br>• Create a timetable with short breaks.<br>• Manage stress through avoiding procrastination, adequate sleep, relaxation and focusing on one task at a time. | 1 mark: goal setting; 1 mark: prioritisation/task breakdown; 1 mark: timetable/time management; 1 mark: practical stress-management strategy.<br><br>Maximum: 4 marks |

| Q. | QUESTION / SUGGESTED VALUE POINTS | MARKING GUIDANCE |
|---|---|---|
| 3 | Your friend receives an email asking for their password and personal information through an unfamiliar link. The message claims that their account will be blocked if they do not respond immediately. What should your friend do, and which safe ICT practices should they follow?<br><br>Suggested value points:<br>• Recognise it as likely phishing.<br>• Do not click the unfamiliar link or provide information.<br>• Verify through the organisation's official website/contact method.<br>• Use strong unique passwords and two-factor authentication.<br>• Report/delete suspicious email and check account activity for unauthorised access. | 1 mark: identifies phishing/risk; 1 mark: avoids link/data disclosure and verifies independently; 1 mark: recommends secure account practices; 1 mark: reports/deletes and/or checks account activity.<br><br>Maximum: 4 marks |

| Q. | QUESTION / SUGGESTED VALUE POINTS | MARKING GUIDANCE |
|---|---|---|
| 4 | A group of students notices that their school produces a large amount of plastic waste every day. They want to develop a small business that addresses this problem. Explain how they could convert this problem into an entrepreneurial opportunity while promoting environmentally responsible practices.<br><br>Suggested value points:<br>• Identify plastic waste as a business opportunity.<br>• Collect, sort and recycle plastic or create useful products from waste.<br>• Sell reusable/recycled products to students/local community.<br>• Conduct a simple market survey and calculate costs/prices.<br>• Promote environmental benefits, creating income while reducing waste. | 1 mark: identifies the problem as an opportunity; 1 mark: proposes a viable recycled/reusable product or service; 1 mark: includes market/cost/pricing or customer consideration; 1 mark: links business activity to environmental benefit.<br><br>Maximum: 4 marks |

| Q. | QUESTION / SUGGESTED VALUE POINTS | MARKING GUIDANCE |
|---|---|---|
| 5 | A student develops an AI-based system to identify students who may require additional academic support. Its results are represented by a confusion matrix. Calculate the accuracy of the model and explain what the result tells you about its performance.<br><br>Suggested value points:<br>• TP = 40; FP = 10; FN = 5; TN = 45.<br>• Total predictions = 100.<br>• Accuracy = (TP + TN) / Total x 100.<br>• Accuracy = (40 + 45) / 100 x 100 = 85%.<br>• The model correctly classified 85 out of 100 students; the remaining 15 predictions were incorrect and should be examined further. | 1 mark: correctly identifies TP/TN (or equivalent matrix values); 1 mark: correct formula/total; 1 mark: correct 85% calculation; 1 mark: correct interpretation.<br><br>Maximum: 4 marks |

# MARK ALLOCATION & EVALUATION CHECKLIST

| Section | Questions Attempted | Marks per Question | Maximum |
|---|---|---|---|
| A – Objective | 4 + 5 + 5 + 5 + 5 | 1 | 24 |
| B-I – Employability | 3 of 5 | 2 | 6 |
| B-II – AI Evaluation & Data | 4 of 6 | 2 | 8 |
| B-III – Applied / Case-Based | 3 of 5 | 4 | 12 |
| **TOTAL** | **15 questions** | — | 50 |

☐ Every attempted question/part has been assessed.

☐ Objective answers have been awarded only for the correct option.

☐ Subjective answers have been assessed using value points and equivalent correct responses.

☐ Numerical working and final interpretation have been checked where applicable.

☐ No question has been awarded more than its prescribed maximum.

☐ Section-wise and grand totals have been checked.